HDOC Day-47

AIM

To write a C program to find the leftmost pivot index of a variable-length integer array using pointers and a user-defined function named **pivot_index()**.

PROBLEM STATEMENT

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named **pivot_index()** using its address. Use pointers to pass the array between the calling function and pivot_index(). The function must find the leftmost index where the sum of elements strictly to the left is equal to the sum of elements strictly to the right. For the first element, the left sum is 0, and for the last element, the right sum is 0. Display -1 if no pivot index exists.

ALGORITHM

1. Start the program.
2. Read the size **n** of the array.
3. Read **n** integer elements into the variable-length array.
4. Pass the address of the array and its size to **pivot_index()**.
5. Find the total sum of all array elements.
6. Initialize **left_sum = 0**.
7. For each index i, calculate **right_sum = total_sum - left_sum - arr[i]**.
8. If **left_sum == right_sum**, return i immediately so the leftmost pivot index is obtained.
9. Otherwise add arr[i] to left_sum and continue.
10. If no pivot index is found, return -1.
11. Display the pivot index or -1.
12. Stop the program.

C PROGRAM

```c
#include <stdio.h>

int pivot_index(int *arr, int n)
{
    int total_sum = 0;
    int left_sum = 0;
    int right_sum;
    int i;

    for (i = 0; i < n; i++)
        total_sum = total_sum + *(arr + i);

    for (i = 0; i < n; i++)
    {
        right_sum = total_sum - left_sum - *(arr + i);

        if (left_sum == right_sum)
            return i;

        left_sum = left_sum + *(arr + i);
    }

    return -1;
}

int main()
{
    int n, i, index;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    index = pivot_index(arr, n);

    if (index == -1)
        printf("-1\n");
    else
        printf("Pivot index: %d\n", index);

    return 0;
}
```

TEST CASE TABLE

Test Case	Array	Expected Output
1	1 7 3 6 5 6	Pivot index: 3
2	1 2 3	-1
3	2 1 -1	Pivot index: 0

OUTPUT

Test Case 1

```
Enter the size of array: 6
Enter 6 elements: 1 7 3 6 5 6
Pivot index: 3
```

Test Case 2

```
Enter the size of array: 3
Enter 3 elements: 1 2 3
-1
```

Test Case 3

```
Enter the size of array: 3
Enter 3 elements: 2 1 -1
Pivot index: 0
```

RESULT

The program successfully finds and displays the leftmost pivot index where the left sum is equal to the right sum. If no pivot index exists, it displays -1.